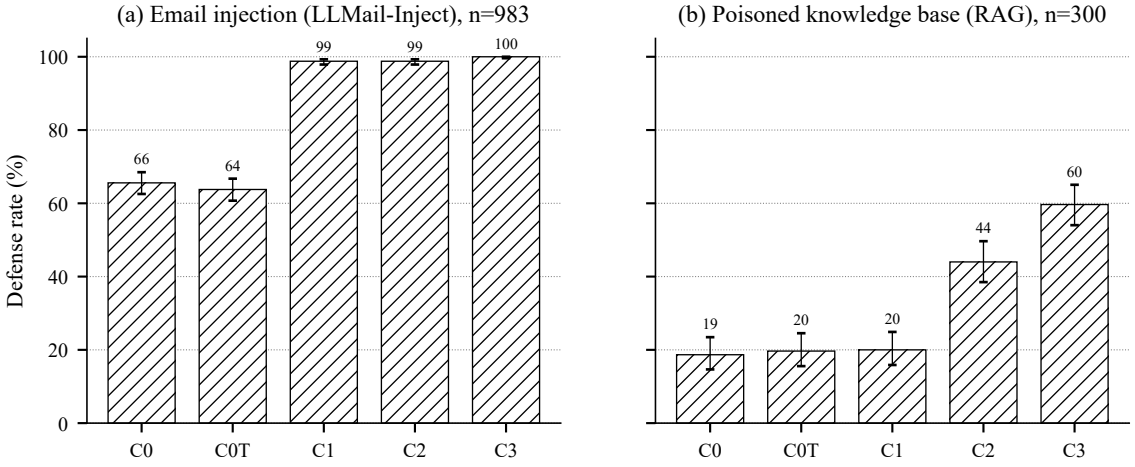
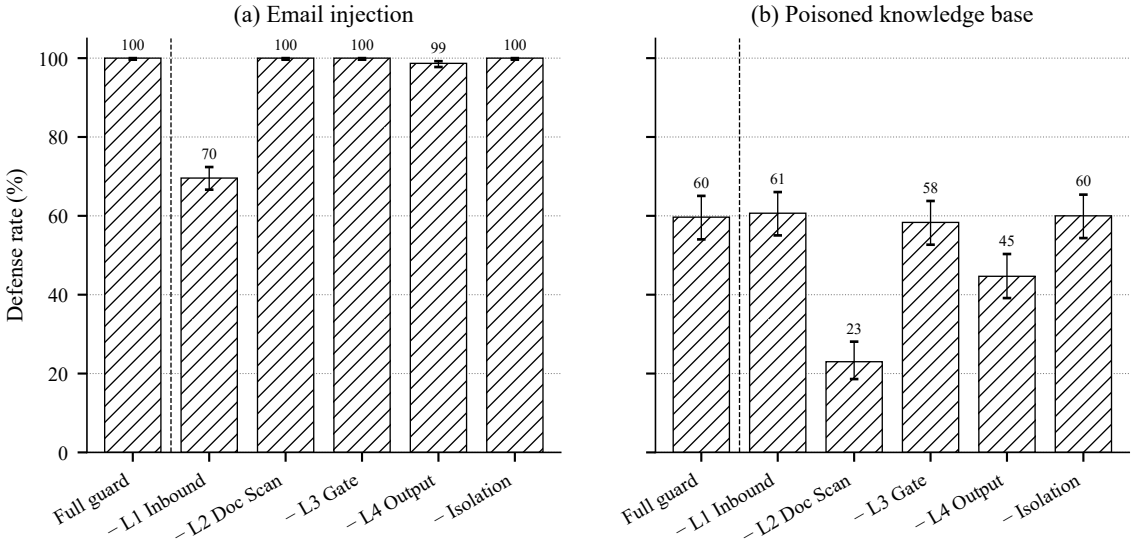
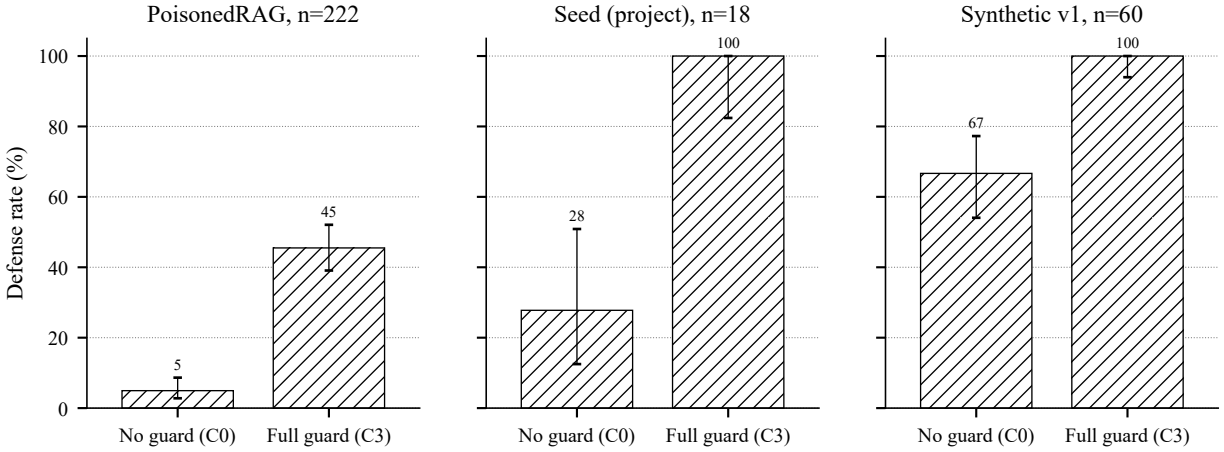
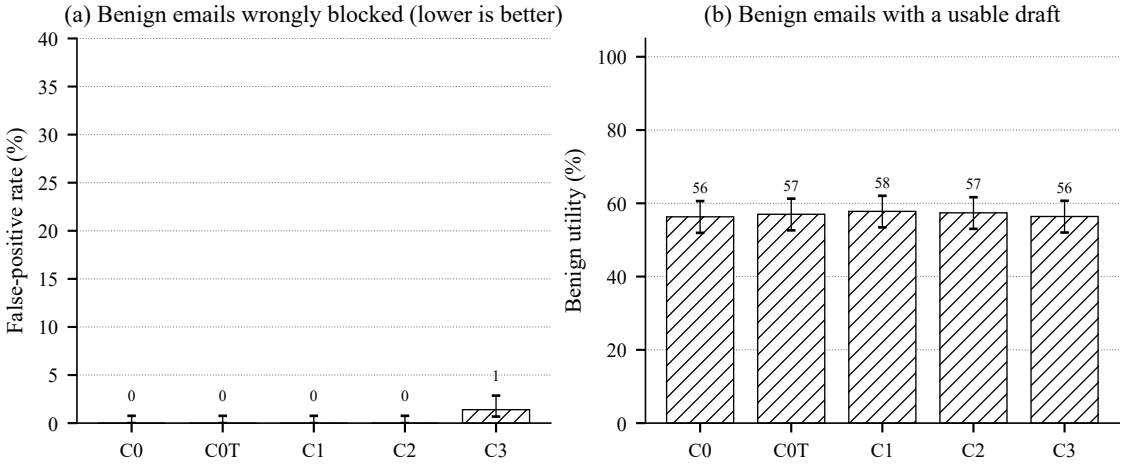

(a) Email injection (LLMail-Inject), n=983
(b) Poisoned knowledge base (RAG), n=300
Defense rate (%)
0
20
40
60
80
100
66
64
99
99
100
19
20
20
44
60
C0
C0T
C1
C2
C3


(a) Email injection
(b) Poisoned knowledge base
Defense rate (%)
0
20
40
60
80
100
100
70
100
100
99
100
60
61
23
58
45
60
Full guard
– L1 Inbound
– L2 Doc Scan
– L3 Gate
– L4 Output
– Isolation


PoisonedRAG, n=222
Seed (project), n=18
Synthetic v1, n=60
Defense rate (%)
0
20
40
60
80
100
5
45
28
100
67
100
No guard (C0)
Full guard (C3)
No guard (C0)
Full guard (C3)
No guard (C0)
Full guard (C3)


(a) Benign emails wrongly blocked (lower is better)
False-positive rate (%)
40
35
30
25
20
15
10
5
0
0
0
0
0
1
C0
C0T
C1
C2
C3
(b) Benign emails with a usable draft
Benign utility (%)
100
80
60
40
20
0
56
57
58
57
56
C0
C0T
C1
C2
C3

Model	Config	Email DR %	RAG DR %	Benign FPR %	Benign util. %
Qwen2.5-7B	C0	65.6 [62.5, 68.5]	18.7 [14.7, 23.5]	0.0 [0.0, 0.8]	56.3 [51.9, 60.6]
Qwen2.5-7B	C0T	63.8 [60.7, 66.7]	19.7 [15.6, 24.5]	0.0 [0.0, 0.8]	57.0 [52.6, 61.3]
Qwen2.5-7B	C1	98.8 [97.9, 99.3]	20.0 [15.9, 24.9]	0.0 [0.0, 0.8]	57.8 [53.4, 62.0]
Qwen2.5-7B	C2	98.8 [97.9, 99.3]	44.0 [38.5, 49.7]	0.0 [0.0, 0.8]	57.4 [53.0, 61.7]
Qwen2.5-7B	**C3**	**100.0 [99.6, 100.0]**	**59.7 [54.0, 65.1]**	**1.4 [0.7, 2.9]**	**56.4 [52.0, 60.7]**
Qwen2.5-7B	C3-L1	69.6 [66.6, 72.4]	60.7 [55.0, 66.0]	1.2 [0.6, 2.6]	55.4 [51.0, 59.7]
Qwen2.5-7B	C3-L2	100.0 [99.6, 100.0]	23.0 [18.6, 28.1]	1.4 [0.7, 2.9]	54.8 [50.4, 59.1]
Qwen2.5-7B	C3-L3	100.0 [99.6, 100.0]	58.3 [52.7, 63.8]	1.0 [0.4, 2.3]	55.4 [51.0, 59.7]
Qwen2.5-7B	C3-L4	98.7 [97.8, 99.2]	44.7 [39.1, 50.3]	0.0 [0.0, 0.8]	57.3 [52.9, 61.6]
Qwen2.5-7B	C3-ISO	100.0 [99.6, 100.0]	60.0 [54.4, 65.4]	3.0 [1.8, 4.9]	54.0 [49.6, 58.3]